Vaishaly Verma

Research scholar
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Personal Details

- **Full Name:** Vaishaly Verma
- **Father's Name:** Late Ramu Verma
- **Nationality:** Indian
- **Date of Birth:** July 7, 1994
- **Address:** H.No. 170, Block 13, Govind Nagar, Kanpur Nagar, Uttar Pradesh – 208006, India
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About Myself

- I am a passionate mathematics researcher with over five years of experience in the field of optimization. My research primarily focuses on fuzzy linear fractional programming problems, with particular emphasis on developing efficient optimization techniques under uncertainty. I have worked extensively with generalized fuzzy parameters to model and solve complex multi-objective fractional optimization problems.
- I have recently submitted my Ph.D. thesis in mathematics. I have an established publication record in reputed peer-reviewed journals, with significant research contributions in optimization. Furthermore, I have actively served as a reviewer for recognized academic journals and have reviewed several research papers.

Research Interests

- Optimization Theory
- Fuzzy Fractional Programming
- Intuitionistic Fuzzy Optimization
- Neutrosophic Optimization
- Multi-Objective Optimization
- Evolutionary Algorithms

Education

2021 – Present

- **Ph.D. in Mathematics**, Motilal Nehru National Institute of Technology (MNNIT), Allahabad (Prayagraj), Uttar Pradesh, India
  Thesis title: *A Study on Generalized Fuzzy Parameter-Based Linear Fractional Programming Problems* **Coursework grade: 9/10**
  Supervisor: Prof. Pitam Singh, Department of Mathematics, MNNIT Allahabad

Education (continued)

2016 – 2018 **M.Sc. in Mathematics**, Department of Mathematics, Indian Institute of Technology (IIT) Patna, Bihta, Patna, Bihar, India (801106) **CGPA: 7.2/10**
Thesis: *Control Theory*
Supervisor: Dr. Nutan Kumar Tomar, Department of Mathematics, IIT Patna

2012 – 2015 **B.Sc. in Mathematics**, D.B.S college, Chhatrapati Shahu Ji Maharaj University (C.S.J.M. University), Kanpur, Uttar Pradesh, India

National Level Examinations

2021 **GATE (Mathematics)**: Score 417, All India Rank 1141.

GATE (Statistics): Score 173, All India Rank 182.

2016 **JAM (Mathematics)**: All India Rank 1468.

Research Publications

Published

- Vaishaly Verma and Pitam Singh, "Neutrosophic Linear Fractional Optimization Problem Using a Fuzzy Goal Programming Approach," *International Journal of Modeling, Simulation, and Scientific Computing*, vol. 15, no. 03, p. 2450027, 2024.
- Vaishaly Verma and Pitam Singh, "A Survey on Generalized Fuzzy Parameter-Based Fractional Programming Problems," *New Mathematics and Natural Computation*, vol. 21, no. 01, pp. 281–321, 2025.

Communicated

- Vaishaly Verma and Pitam Singh, "An Approach for Optimizing Fully Fuzzy Multi-Objective Linear Fractional Programming Problems with an Application in Production Planning."
- Vaishaly Verma and Pitam Singh, "A Fuzzy Harmonic Mean-Based Method for Solving Intuitionistic Fuzzy Multi-Objective Linear Fractional Optimization Problems and Its Application in Organizational Planning."
- Vaishaly Verma and Pitam Singh, "Solving Multi-Objective Fractional Transportation Problems in a Multi-Choice Neutrosophic Environment with Stochastic Constraints."

Conferences/Workshops

Conferences

- International Conference on Optimization with Artificial Intelligence and Soft Computing (ICOAISC-2024), organized by the Department of Science and Humanities, National Institute of Technology Nagaland, September 30–October 01, 2024.
- International Conference on Computational Operations Research and Algorithmic Game Theory, organized by the Department of Mathematics, Indian Statistical Institute, Delhi, January 21–23, 2025.
- International Conference on Nonlinear Optimization and Applications (ICNOA 2025), organized by the Faculty of Technology and the Department of Mathematics, Banaras Hindu University, October 29–31, 2025.
- International Conference on Mathematical Sciences and Computational Intelligence (ICMSCI), organized by the Department of Mathematics, P.G.D.A.V. College, University of Delhi, Indian Institute of Technology (IIT), Mandi & National Institute of Technology (NIT), Uttarakhand. February 20–22, 2026.

Conferences/Workshops (continued)

Workshops

- International Symposium on Optimization Theory and its Applications (ISOTA-2023), organized by the Department of Mathematics, Thapar Institute of Engineering and Technology, April 13–15, 2023.
- Workshop on "Mastering Optimization: Machine Learning Models and Real-World Applications", organized by the Department of Mathematics, VIT Bhopal University, October 10–12, 2024.
- Summer School on "Evolutionary Algorithms and Machine Learning: Theory and Applications", organized by the Department of Mathematics & Computing, Dr. B. R. Ambedkar National Institute of Technology, Jalandhar, June 16–20, 2025.
- Online FDP on "Emerging Trends in Fuzzy Mathematics", organized by the School of Advanced Sciences (SAS), Vellore Institute of Technology, Chennai, June 28–30, 2025.
- Two-Day International Workshop on "AI Tools for Research and Academic Writing", organized by Manuscriptpedia, Kanniyakumari, January 24–25, 2026 (online mode).

Teaching Experience

Teaching Assistant (TA), Motilal Nehru National Institute of Technology (MNNIT), Allahabad

- Served as a Teaching Assistant for undergraduate engineering courses during the Ph.D. tenure.
- Conducted laboratory sessions in C Programming, providing hands-on guidance and evaluation.
- Assisted in mentoring students and clarifying programming and computational concepts.

Skills

Languages	**English:** Strong reading, writing, and speaking proficiency.
	Hindi: Strong reading, writing, and speaking proficiency.
Computational Software	MATLAB, LINGO, C Programming …
Text Processors	LaTeX, MS PowerPoint, MS Excel, MS Word, …
Communication	Participated in several national and international conferences, webinars, and workshops. Received a **Best Oral Presentation Award** for outstanding research presentation. Engagement with diverse academic audiences across various scientific disciplines has significantly enhanced my communication, presentation, and subject knowledge.

Extracurricular Activities

- Interests include sports, cultural activities, and creative writing.
- Won three medals (including gold and silver) in Shot Put and Discus Throw at the institutional level.
- Represented the institute in Shot Put and Discus Throw at the prestigious Inter-IIT Sports Meet.

References

Prof. Pitam Singh
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Prof. Nutan Kumar Tomar
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